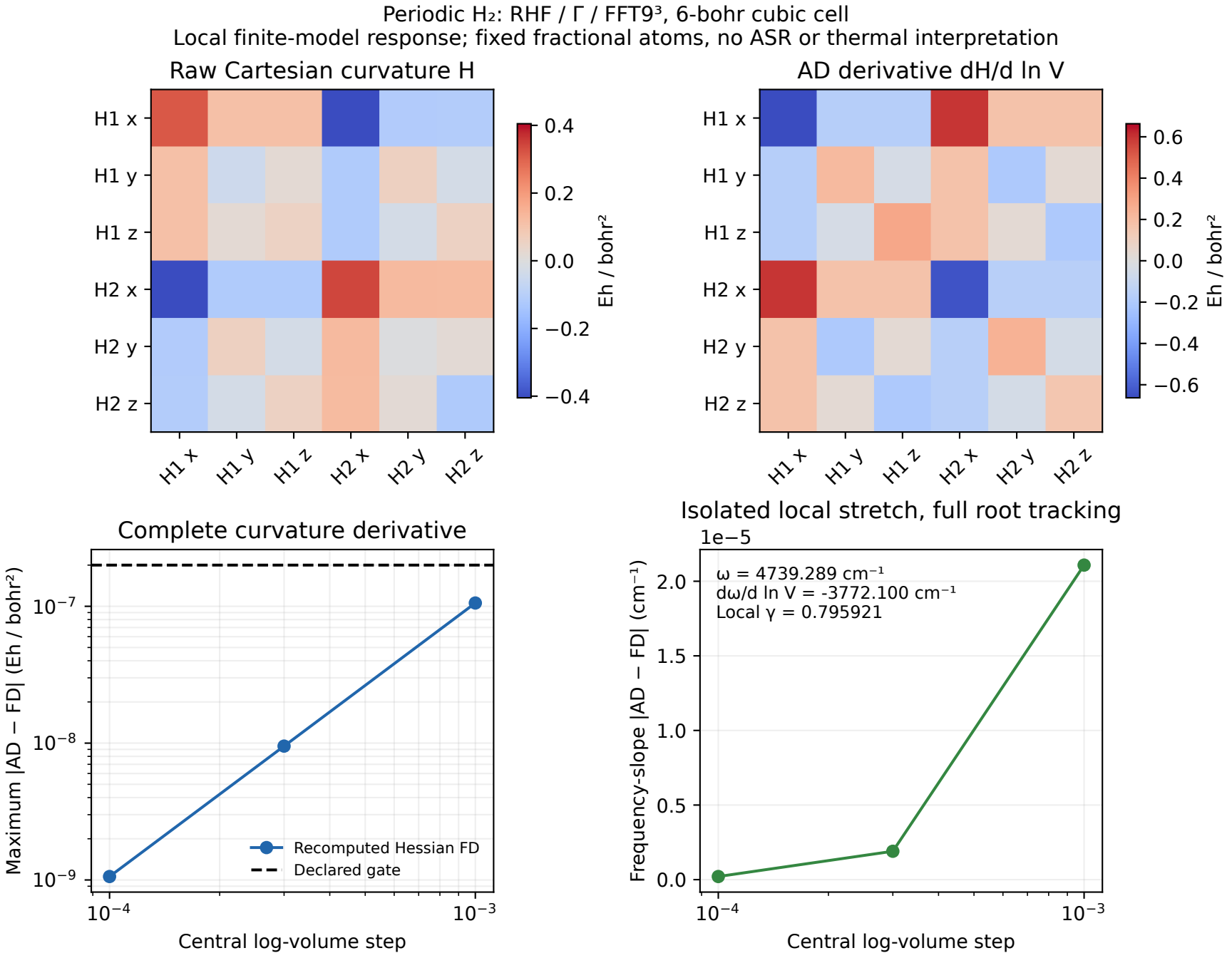
Periodic H₂: RHF / Γ / FFT9³, 6-bohr cubic cell
Local finite-model response; fixed fractional atoms, no ASR or thermal interpretation
Raw Cartesian curvature H
H1 x
H1 y
H1 z
H2 x
H2 y
H2 z
0.4
0.2
0.0
−0.2
−0.4
Eh / bohr²
AD derivative dH/d ln V
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Complete curvature derivative
Maximum |AD − FD| (Eh / bohr²)
10^{-7}
10^{-8}
10^{-9}
Recomputed Hessian FD
Declared gate
10^{-4}
10^{-3}
Central log-volume step
Isolated local stretch, full root tracking
1e−5
Frequency-slope |AD − FD| (cm⁻¹)
2.0
1.5
1.0
0.5
0.0
ω = 4739.289 cm⁻¹
dω/d ln V = -3772.100 cm⁻¹
Local γ = 0.795921
10^{-4}
10^{-3}
Central log-volume step